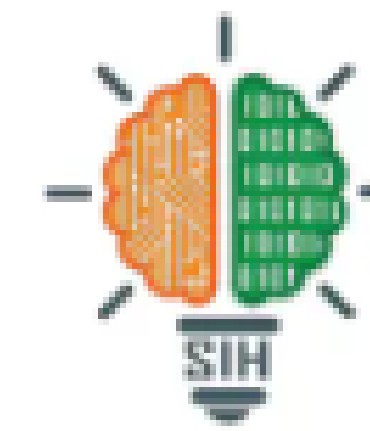


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- **Problem Statement ID** – SIH1627
- **Problem Statement Title**- Drug Inventory and supply chain Tracking system
- **Theme**- MedTech / BioTech / HealthTech
- **PS Category**- Software
- **Team ID**- 161
- **Team Name** - Laksh



❖ Proposed Solution

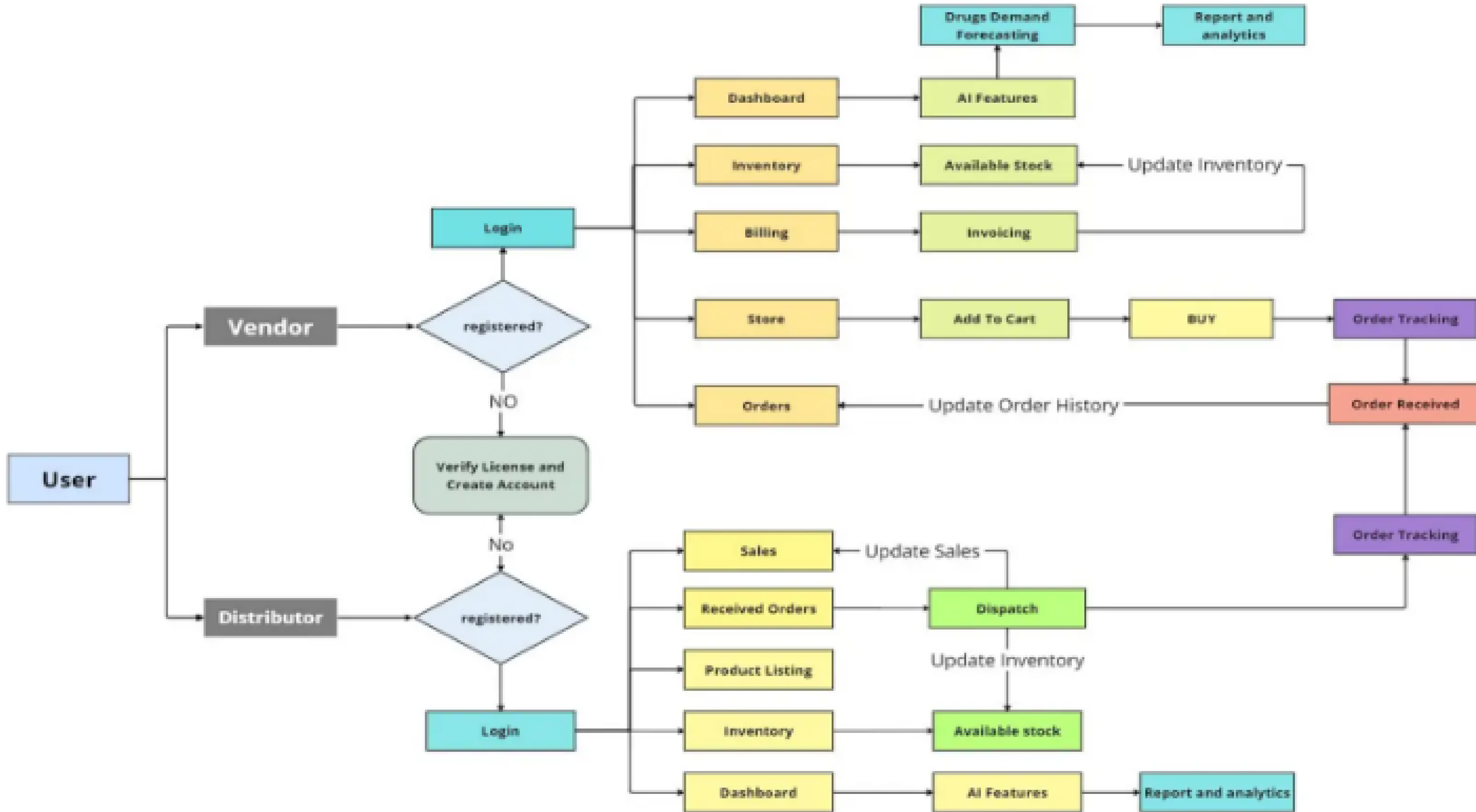
- The proposed solution is a Drug Inventory and Supply Chain Tracking System designed to **streamline the distribution**.
- Availability of drugs in hospitals and medical institutions.
- The system will use modern technologies and **Artificial Intelligence** to provide **real-time tracking, monitoring, and management** of drug inventory across the supply chain.
- The system aims to ensure that the **right drugs** are available in the **right quantity**, at the **right time, place, and cost**, in the **right condition**, and for the **right people**.

How the Solution Addresses the Problem:

- Ensures Drug Availability
- Improves Procurement Efficiency
- Monitors Drug Consumption Patterns
- Reduces Costs and Waste
- Improves Data Security

Innovation and Uniqueness of the Solution:

- Integrated Multi-Database Approach
- Real-Time Dashboard Monitoring
- Focus on Data-Driven Decisions
- Enhanced Security with Modern Protocols
- AI based Forecasting



Feasibility:	Challenges and Risks:	Strategies for Overcoming The Challenges:
1. Technical Feasibility - Technology Stack - Cloud Integration - Security 2. Operational Feasibility - Ease of Use - User Adoption 3. Financial Feasibility - Cost Considerations - Scalability	1. Data Accuracy and Integrity - Challenge: Ensuring data is up to date. - Risk: Errors can cause inventory issues. 2. System Integration - Challenge: Connecting with existing systems. - Risk: May loss data . 3. Cybersecurity - Challenge: Protecting sensitive data. - Risk: Breaches can cause theft and losses.	- Ensuring Data Accuracy and Integrity - Smooth Integration with Existing Systems - Enhancing Cybersecurity Measures - Managing Vendor Reliability - Ensuring Regulatory Compliance

Benefits:

1. Improved Drug Availability:

Ensures essential medicines are always available, reducing shortages.

2. Cost Reduction:

Optimizes procurement and reduces wastage, leading to cost savings for hospitals and institutions.

3. Real-Time Monitoring:

Enables real-time tracking of inventory, shipments, and consumption patterns through a dashboard.

4. Vendor Accountability:

Improves vendor performance by tracking their activities and ensuring timely deliveries.

5. Enhanced Decision-Making:

Provides data-driven insights into drug usage, improving procurement planning.

Impacts:

1. Social Impact:

Better access to medicines, improving public health.

2. Environmental Impact:

Reduces pharmaceutical waste and lowers carbon emissions due to optimized transportation and procurement.

3. Economic Impact:

Lowers healthcare costs for patients and increases competition among vendors, leading to more affordable drug prices.

4. Governance and Policy Impact:

Provides governments with data to make informed decisions on healthcare policy and resource allocation.

5. Job Creation and Innovation:

Creates jobs in IT, supply chain management, and logistics, and fosters innovation in pharmaceutical distribution.



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3. **“A Novel Framework for Pharmaceutical Supply Chain Management using Distributed Ledger and Smart Contracts”**, Sandip Jangir, Ajit Muzumdar, Alok Jaiswal, Chirag N. Modi, Sheetal Chandel, C. Vyjayanthi, 2019 10th International Conference on Computing, Communication and Networking Technologies (ICCCNT)
4. **“A Survey on Supply Chain Security: Application Areas, Security Threats, and Solution Architectures”**, Vikas Hassija, Vinay Chamola, Vatsal Gupta, Sarthak Jain, Nadra Guizani, IEEE Internet of Things Journal (Volume: 8, Issue: 8, 15 April 2021)
5. **“Data analytics in pharmaceutical supply chains: state of the art, opportunities, and challenges”**, Angie Nguyen, Samir Lamouri, Robert Pellerin, Simon Tamayo, Béranger Lekens, Received 30 Oct 2020, Accepted 16 Jun 2021, Published online: 19 Jul 2021